

Differences in Differences

PS200C, Spring 2026

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Roadmap

- 1 Theory: Potential outcomes, identification, key quantities
- 2 Randomization
 - difference in means, variance estimation
 - covariate adjustment
 - blocking
 - cluster randomization
- 3 Selection on Observables
 - sub-classification
 - matching (on X)
 - weighting (on X)
 - matching and weighting on $Pr(D = 1)$ (propensity scores)
 - regression
 - sensitivity
- 4 Difference in Differences
- 5 Instrumental Variables
- 6 Regression Discontinuity
- 7 More panel methods (time permitting)

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- A (relatively small) number of identification strategies have gained popularity as alternatives: DID, IV, RDD, and some time-series approaches

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- For any of these, you'll find there is some degree of inappropriate faith in these techniques because of this association
- **RESIST!** Understand and critique the assumptions.

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DID in the Time of Cholera

- Prevailing 19th century wisdom was that cholera was spread by “bad air”
- John Snow, a London physician thought otherwise
- In 1849, the Southwark and Vauxhall water company and the Lambeth company got water from the same polluted part of the Thames in central London. Snow assessed cholera mortality in both areas.
- In 1852, Lambeth moved its waterworks up stream.
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Intuitions:

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- Why not just compare the two in 1852?
- Why not just compare Lambeth's service area post-minus-pre?

Two Groups, Two Periods

Definition

Two groups:

- $D = 1$ for (eventually) treated units
- $D = 0$ for control units

Two periods:

- $T = 0$ Pre-Treatment period
- $T = 1$ Post-Treatment period

Potential outcomes $Y_{it}(d)$

- $Y_{it}(1)$: outcome unit i would attain in period t if treated by then
- $Y_{it}(0)$: outcome unit i would attain in period t if not treated

Two Groups and Two Periods

Definition

Causal effect for unit i at time t is

$$\tau_{it} = Y_{it}(1) - Y_{it}(0)$$

Observed outcomes Y_{it} are realized as

$$Y_{it} = Y_{it}(1) \cdot (D_i \cdot (t = 1)) + Y_{it}(0) \cdot (1 - D_i \cdot (t = 1))$$

Estimand (ATT post-treatment)

$$\tau_{ATT} = \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1]$$

Two Groups and Two Periods

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Observed Moments:

	Post-Period (T=1)	Pre-Period (T=0)
Treated D=1	$\mathbb{E}[Y_1(1) D = 1]$	$\mathbb{E}[Y_0(0) D = 1]$
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Problem

Missing potential outcome: $\mathbb{E}[Y_1(0)|D = 1]$, ie. what is the average non-treatment outcome in the post-treatment period, for the treated?

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ID Strategy 1 (not DID): Over-time (“Post-minus-Pre”) Comparison

- Estimand: $\mathbb{E}[Y_1(1)|D = 1] - \mathbb{E}[Y_0(0)|D = 1]$
- Works if: $\mathbb{E}[Y_0(0)|D = 1] = \mathbb{E}[Y_1(0)|D = 1]$
- What does this assumption mean?

Two Groups and Two Periods

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ID Strategy 2 (not DID): Cross-Sectional Comparison (in Post)

- Use: $\mathbb{E}[Y_1|D = 1] - \mathbb{E}[Y_1|D = 0]$
- Assumes: $\mathbb{E}[Y_1(0)|D = 0] = \mathbb{E}[Y_1(0)|D = 1]$
- What does this mean?

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ID Strategy 3: Difference-in-Differences (DID)

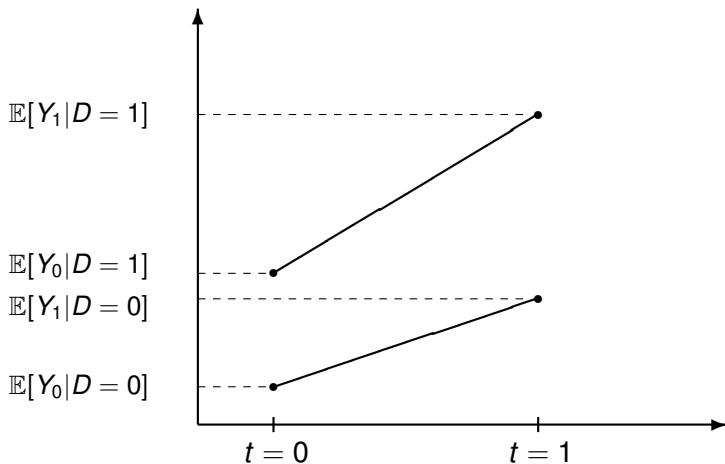
- Use:

$$\mathbb{E}[Y_1 - Y_0|D = 1] - \mathbb{E}[Y_1 - Y_0|D = 0]$$

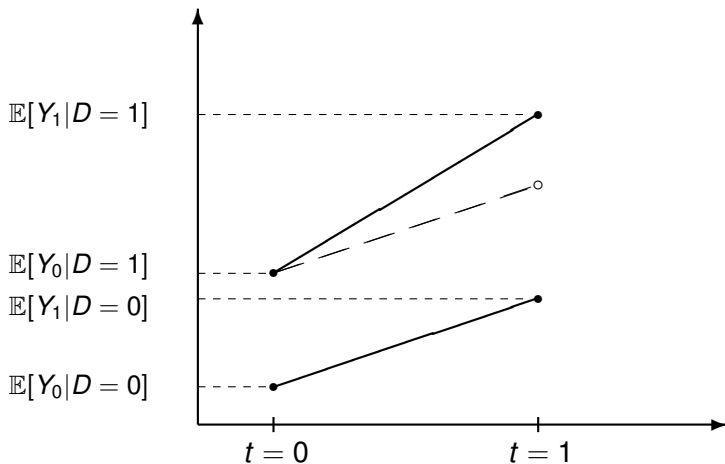
- Assumes: $\mathbb{E}[Y_1(0) - Y_0(0)|D = 1] = \mathbb{E}[Y_1(0) - Y_0(0)|D = 0]$

“Parallel trends” assumption

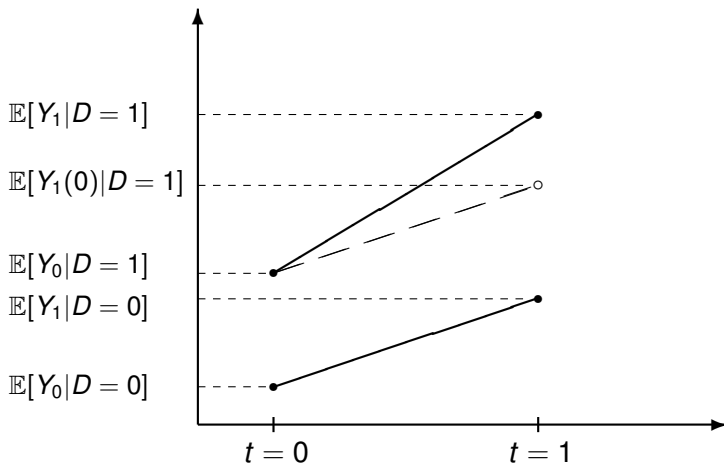
Graphical Representation: Difference-in-Differences



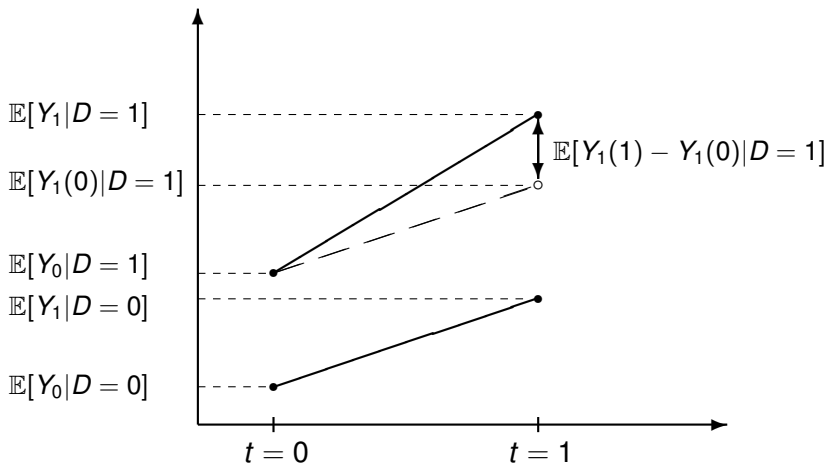
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Identification with Difference-in-Differences

Identification Assumption (Parallel Trends)

Conventionally we write,

$$\mathbb{E}[Y_1(0) - Y_0(0)|D = 1] = \mathbb{E}[Y_1(0) - Y_0(0)|D = 0]$$

But note that this is equivalent to:

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Identification Result

Under parallel trends, the ATT is identified by the DID estimator:

$$\begin{aligned}\tau_{DID} &= \{\mathbb{E}[Y_1|D = 1] - \mathbb{E}[Y_0|D = 1]\} - \{\mathbb{E}[Y_1|D = 0] - \mathbb{E}[Y_0|D = 0]\} \\ &= \mathbb{E}[Y_1(1) - Y_1(0)|D = 1] = ATT\end{aligned}$$

Proof

$$\tau_{DID} = \{\mathbb{E}[Y_1|D = 1] - \mathbb{E}[Y_0|D = 1]\} - \{\mathbb{E}[Y_1|D = 0] - \mathbb{E}[Y_0|D = 0]\}$$

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- $DID = \mathbb{E}[\Delta Y | D=1] - \mathbb{E}[\Delta Y | D=0] = (\delta_1 + \tau_{ATT}) - \delta_0$.
- ...equals τ_{ATT} if $\delta_0 = \delta_1$ (parallel trends).

Why be skeptical of parallel trends?

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Parallel trends isn't a *weaker* assumption—it's a *different* one. You need a story for why treatment assignment and unit-level drift are unrelated.

Ex ante reasons for hope or worry

Hopeful settings. Parallel trends is more plausible when the *selection mechanism* is blind to $Y(0)$ dynamics:

- **Decision lag:** treatment was decided well before it kicked in, weakening the relationship between who is treated and the trend come treatment time.

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Worrisome settings. Something *changing abruptly* around treatment time that shapes *both* who gets treated and the outcome:

- Treatment is a *response* to a recent shock or drift (Ashenfelter dip; programs target recent decliners).
- Latent forces influence both treatment taking (timing) and how $Y(0)$ would change absent treatment.

Difference-in-Differences: Estimators

Estimand (DID)

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Estimator

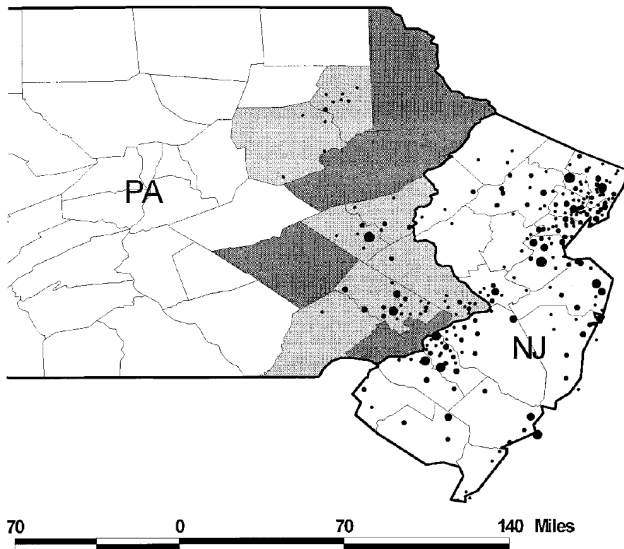
$$\frac{1}{N_1} \sum_{D_i=1} \{Y_{i1} - Y_{i0}\} - \frac{1}{N_0} \sum_{D_i=0} \{Y_{i1} - Y_{i0}\},$$

where N_1 and N_0 are the number of treated and control units respectively.

Example: Minimum wage laws and employment

- Do higher minimum wages decrease low-wage employment?
- Card and Krueger (1994) consider impact of New Jersey's 1992 minimum wage increase from \$4.25 to \$5.05 per hour
- Compare employment in 410 fast-food restaurants in New Jersey and eastern Pennsylvania before and after the rise
- Survey data on wages and employment from two waves:
 - Wave 1: March 1992, one month before min. wage increase
 - Wave 2: December 1992, eight months after increase

Locations of Restaurants (Card and Krueger 2000)



Sample Means: Minimum wage laws and employment

Variable	Stores by state		
	PA	NJ	Difference,
	(i)	(ii)	NJ – PA (iii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	-2.89 (1.44)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	-0.14 (1.07)
3. Change in mean FTE employment	-2.16 (1.25)	0.59 (0.54)	2.76 (1.36)

Difference-in-Differences: Estimators

Estimator (Regression: Repeated Cross-Sections)

In the two period case, the same estimator can be computed by regression:

$$Y = \mu + \gamma \cdot D + \delta \cdot T + \tau \cdot (D \cdot T) + \varepsilon,$$

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	After (T=1)	Before (T=0)	After - Before
Treated D=1	$\mu + \gamma + \delta + \tau$	$\mu + \gamma$	$\delta + \tau$
Control D=0	$\mu + \delta$	μ	δ
Treated - Control	$\gamma + \tau$	γ	τ

Again, parallel trends: “time-effect” δ same for treated and controls

Regression: Minimum wage laws and employment

```
> d <- read.dta("CK1994_longformat.dta", convert.factors = FALSE)
> head(d[, c('ID', 'nj', 'postperiod', 'emptot')])
```

	ID	nj	postperiod	emptot
1	1	0	0	40.50
2	1	0	1	24.00
3	2	0	0	13.75
4	2	0	1	11.50
5	3	0	0	8.50
6	3	0	1	10.50

Regression: Minimum wage laws and employment

You can reproduce the mean-differences estimator with a regression:

```
> ols <- lm(emptytot ~ postperiod * nj, data = d)
> coeftest(ols)
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	23.3312	1.0719	21.7668	< 2e-16	***
postperiod	-2.1656	1.5159	-1.4286	0.15351	
nj	-2.8918	1.1935	-2.4229	0.01562	*
postperiod:nj	2.7536	1.6884	1.6309	0.10331	

Three famous DiD studies: is parallel trends credible?

For each: imagine a *time-varying* confounder. What might make $\Delta Y(0)$ differ between treated and control?

- 1 **Card & Krueger (1994)**: NJ raises its minimum wage; PA doesn't.
 Y = fast-food employment. Compare NJ vs PA, before/after.

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- 4 **Snow's cholera study**: Lambeth (moves water works) vs. S&V (does not move water works), look at cholera mortality.

Some sneaky ways parallel trends can be violated

- 1 **Ashenfelter dip / mean reversion:** treatment is a response to a recent bad draw (people enroll in a training program *because* earnings just dipped; NGOs target villages that had a setback). The treated group is poised to recover even without treatment—so $\Delta Y(0)$ differs.

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- 4 **Functional-form dependence:** differencing is not scale-invariant. If outcomes grow at rates that depend on their *levels*, trends won't be parallel even with purely static confounding. Parallel in levels $\neq$ parallel in logs.

Checks for Difference-in-Differences Design

Is the parallel-trends assumption testable?

Checks for Difference-in-Differences Design

Two kinds of **falsification** tests for non-parallel trends

- 1 Falsification test using data for prior periods
 - Given parallel trends during periods $t = -1, 0, 1$, we have:

$$\mathbb{E}[Y_0 - Y_{-1} | D = 1] - \mathbb{E}[Y_0 - Y_{-1} | D = 0] = 0$$

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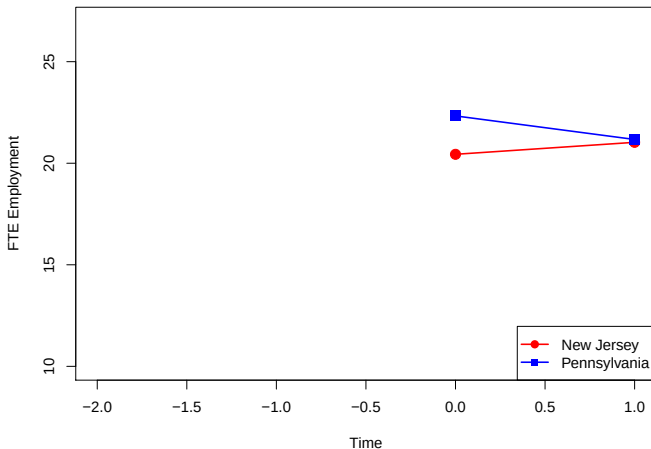
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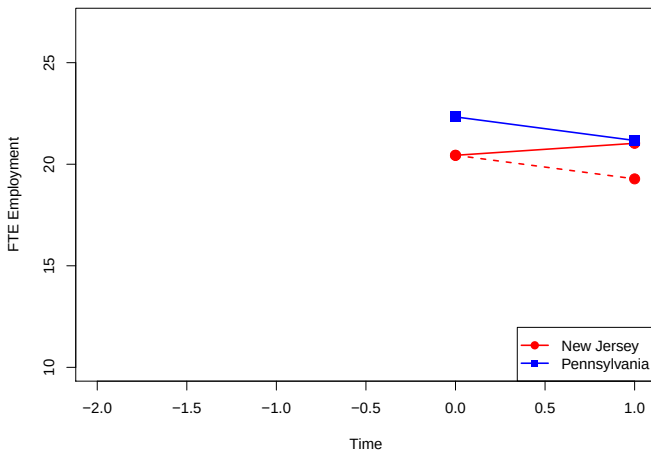
2 Falsification test using alternative outcome or treated sub-group that is not supposed to be affected by the treatment

- troubling if DID from the placebo outcome or un-movable group is non-zero.

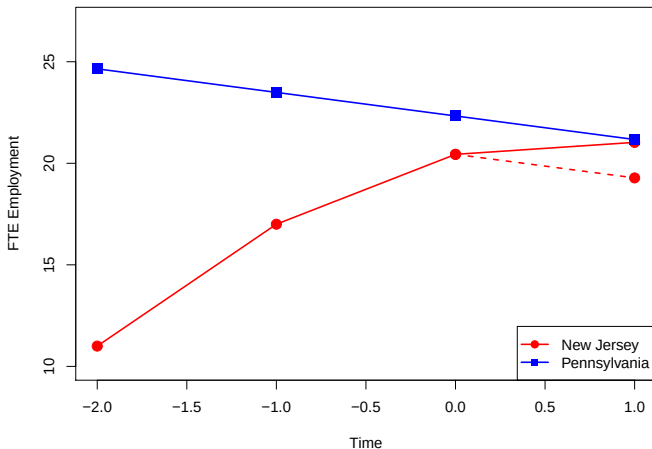
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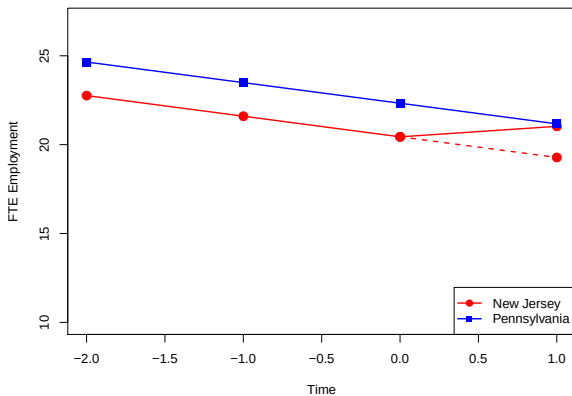


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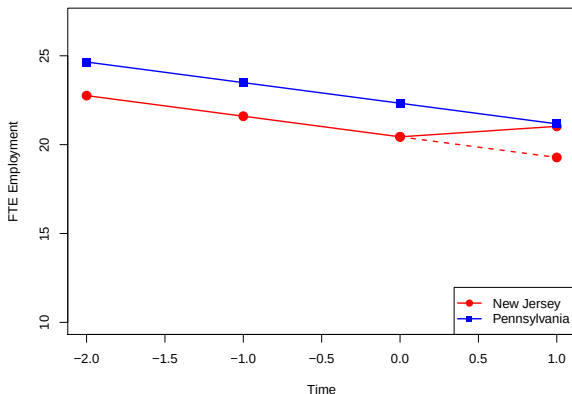


Definitely bad.

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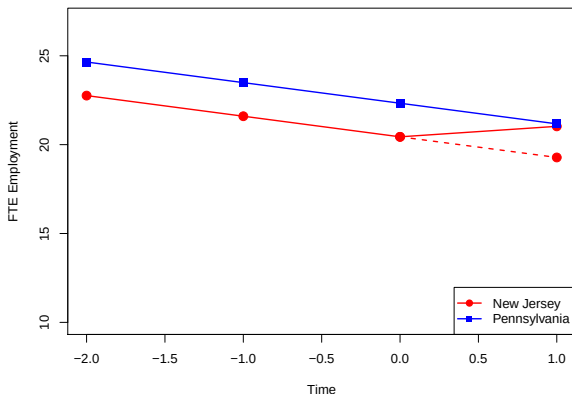


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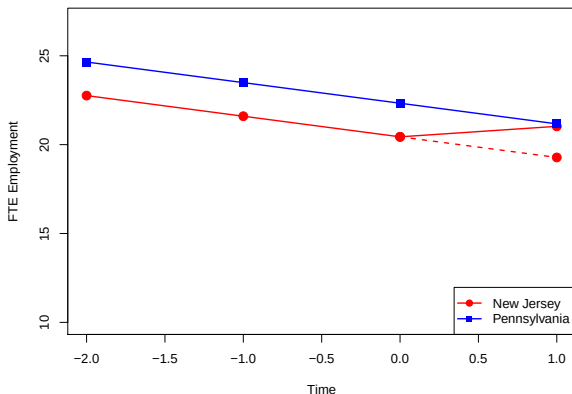
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Parallel pre-trends $\neq$ parallel trends.

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Not definitely bad, but why did they take treatment then?
Parallel pre-trends $\neq$ parallel trends.
You cannot test parallel trends.

Other approaches to worrying about parallel trends

1 Sensitivity/ partial ID

- Best-known is perhaps Rambachan & Roth (2023)
- Suggests checking if the effect estimate can survive non-parallelness k times larger than the non-parallelness in the prior trends.
- A lot of papers don't even survive at $k = 1$, which is clearly bad (Chiu, Lan, Liu, Xu, *APSR* 2025)
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We won't cover as much as I like but there are other ID opportunities for similar settings, and other sensitivity analyses depending on what you are prepared to reason about.

Vote in the comments to affect the story.

“Modern” DID?

- Classic DID above was two groups, two periods. Often you have units moving into treatment at different times (“staggered adoption”), and you’d still like to rely on something like parallel trends.

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 - **Effect heterogeneity** across cohorts or time-since-treatment.
- Intuition: with homogeneous effects, TWFE implicitly “adjusts” the already-treated controls back to their counterfactual by subtracting the one constant τ . Under heterogeneity, no single adjustment works—so contaminated “controls” bleed into the estimate.

Staggered DID: the core idea

It is simple at bottom:

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Simplest fallback: run a clean 2×2 DID for each adoption wave, look at them, average however you want.

Estimators for Staggered DID

The main approaches are:

Approach	What it does
Sun & Abraham (2021)	Event-study estimator using cohort-specific relative-time indicators; avoids contamination from already-treated units; valid under heterogeneous effects and staggered timing.
Callaway & Sant'Anna (2021)	Estimates group-time $ATT(g, t)$ using only not-yet-treated units as controls; flexible aggregation; avoids negative weighting.
Borusyak, Jaravel & Spiess (2021); Liu, Wang & Xu (2022)	Fits $Y(0)$ model with unit + time FEs on <i>untreated cells only</i> , impute $\hat{Y}_{it}(0)$ for treated cells, then aggregate effect estimates explicitly. Essentially g-computation with FE predictors. Implemented in <code>fec</code> .
de Chaisemartin & D'Haultfœuille (2020)	Reweights valid 2×2 DID comparisons with convex weights; interpretable under heterogeneity.

Event-study design: single treatment time

Setup. DGP A: half the units treated at one time, rest never-treated. *Parallel trends holds.* Post-treatment effect is positive, builds slowly.

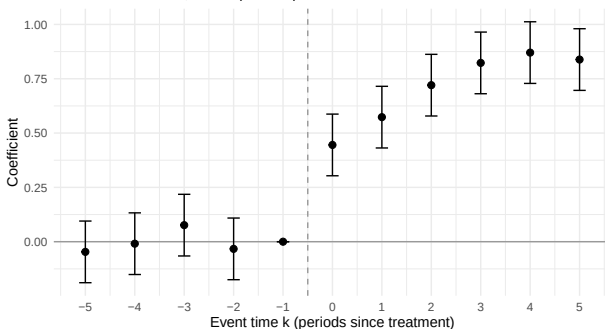
Estimator. TWFE with leads and lags; $k = -1$ omitted as reference:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{k \neq -1} \beta_k \cdot \mathbf{1}\{t - E_i = k\} + \varepsilon_{it}$$

Use it to (a) visualize effect dynamics ($\beta_k, k \geq 0$); (b) check leads ($k < 0$) ≈ 0 —a visual PT diagnostic.

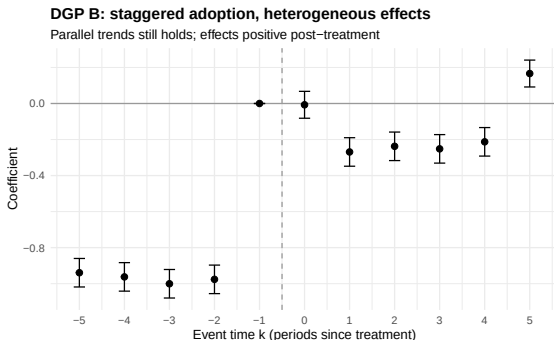
DGP A: single treatment time, homogeneous effect

Parallel trends holds; effects positive post-treatment



Event-study design: staggered adoption

Setup. DGP B: three cohorts adopted at different times; effect heterogeneous across cohorts.
Parallel trends still holds. Effects still positive. **Same TWFE event-study regression** as before.



- Leads appear strongly non-zero—a *false* PT alarm (PT actually holds).
- Lags flip sign, even though every true cohort-time effect is positive.
- Pathology is entirely from the estimator, not the DGP.
- Modern methods (SA, CS, BJS / `fect`) give event-study plots without it.

But remember two things:

Parallel trends is a strong assumption

- If time-varying influences might lead to treatment and changes in “growth”, i.e. $Y_t(0) - Y_{t-1}(0)$, then parallel trends is violated.
- Modern DID methods don't touch this.
- Parallel trends (in the relevant period) is never testable. Don't be fooled by pre-trend tests.